

Homework 1
COMS 3310
Due Thursday Feb 5th at 11:59pm

Late Submission: Friday Feb 6th, 11:59PM (15% Penalty)

Please read the HW guidelines (see next page). Failure to follow the guidelines will result in a lower grade.

Notation: $\mathbb{N}$ is the set of natural numbers, i.e.,

$$\mathbb{N} = \{0, 1, 2, \dots\}.$$

1. Suppose that you have three sets A , B and C which are subsets of some universe.
 - (a) If A and B are disjoint, and $C \subseteq B$, show that if $x \in C$ then $x \notin A$.
 - (b) Show that the sets $A \setminus B$ and $B \setminus C$ are disjoint. (Note that $X \setminus Y$ is the set of elements that are in X but not in Y).
2. Prove that $n \geq 1$, the following equation is correct:

$$\sum_{i=1}^n (i+1)2^i = n2^{(n+1)}$$

Hint: Start by writing out the sum on the left in an “extended” way, without using the Sigma sum notation. That is, write out the first few terms of the sum on the left, and then use “ $+\dots+$ ” and write the final term.

3. A friend tells you that $\langle C, D \rangle$ is an amazing lossless compression/decompression algorithm. A real world example similar to C and D are zip and unzip, or gzip and gunzip. The compression algorithm C allegedly takes any binary string x of length exactly n , and compresses it (makes it shorter) into a string $C(x)$ of length less than n , while the corresponding decompression algorithm D “undoes” the compression (ie makes the compressed string $C(x)$ longer again and recovers the original string x). So your friend claims that for all x of length n , $|C(x)| < |x|$ and $D(C(x)) = x$.
 - (a) If C does not calculate a one-to-one function, what could go wrong?
 - (b) Consider the case $n = 4$. How many binary strings of length 4 are there? How many binary strings of length less than 4 are there?

- (c) In general terms, how many binary strings of length exactly n are there? How many binary strings of length strictly less than n are there?
- (d) Now show that your friend could not possibly be right about the algorithms C and D .
4. A construction company needs to build walls out of standard panels. Each standard panel's width is either 5 feet or 7 feet, and the height of both panel types is a standard height L (the exact value of L is irrelevant but you can think of it as going from floor to ceiling). To build a wall, the panels are lined up side by side in a straight line so that their 5- or 7- foot widths add up to the desired wall length. The height of a wall is always just L . For example, a 12-foot wall can be built by putting a 5-foot panel adjacent to a 7-foot panel. Obviously a 10 foot wall and a 14 foot wall are easy to build out of two panels each. We always use a non-negative whole number of panels of each type.
- (a) What is the largest wall width k which cannot be built out of the five and seven foot panels? How would you prove that a wall of width k cannot be built?
- (b) Prove (using strong induction) that for all natural numbers $n > k$, it is possible to build a wall of width n .
5. Prove that for all natural numbers $n \geq 0$,

$$3 \mid (4^n + 2).$$

Here, " $a \mid b$ " means b "is divisible by" a .

Guidelines:

- Please submit your HW via Gradescope.
- You must handwrite your solutions.
- If your handwriting is not legible, it will not be graded.
- Start the answer to each question on a separate page.
- You are allowed to collaborate with *others*, while doing your homework.
- Definition of *others*: Students who are taking the course this semester.
- You can consult GenAI tools.
- However, you should write the final solutions alone, without consulting anyone. Your writing should demonstrate that you understand the proofs completely.
- For each problem, you should acknowledge all the individuals you consulted and whether you used any GenAI resources. This will not affect your grade. Failure to acknowledge is considered *academic dishonesty*, and it will affect your grade.
- When proofs are required, you should make them both clear and rigorous. Do not hand wave.
- Any regrading requests should be made within one of returning the homework. No requests will be entertained after this one week period.